

Response to Reviewer (Revision Round 6)

Dear Editor and Reviewer,

Thank you for the sixth reading of my manuscript, *A Reproducible Benchmark of Relational Database Access-Layer Performance in Express.js: A Configuration-Specific Comparison on PostgreSQL and MySQL*, and for the clear verdict that the dataset and artifact are publishable while the remaining obstacle is interpretation — the paper claiming more causal specificity and category generality than a single-host, default-configuration design can establish. I have taken that framing seriously and made it the organizing principle of this revision.

I answer below in the reviewer’s own order: the governing change first (the Priority-1 retraction), then the prioritized checklist P1–P10, then the §7 minor points, then the eight questions Q1–Q8. I quote the revised manuscript briefly so every claimed change is verifiable, and I cite sections by name rather than line number, since line numbers drift between builds.

The governing change this round: retracting the same-SQL causal claim (Priority 1)

You were right that the earlier text over-read the same-SQL control. It claimed that the eager-loading *strategy*, not the execution machinery, drove most of the idiomatic gap — a causal decomposition the design cannot support, because collapsing every layer onto identical SQL changes query strategy, query count, prepared-statement behaviour, the raw-versus-ORM API, and result marshalling all at once. I have **withdrawn the causal attribution everywhere and reworded the control as a pure bounding contrast**: both abstracts, the Results same-SQL subsection, the Discussion, and the Conclusion.

The abstract now presents the collapse as a bound, not a mechanism: “the deep/nested fetch spreads $7.15\times$ (PostgreSQL) and $4.96\times$ (MySQL) between native and the slowest ORM, narrowing to $1.68\times$ and $2.01\times$ once every layer runs identical SQL, a compound control that bounds the combined query-strategy-and-formulation contribution without isolating a single mechanism.”

The Results *Same-SQL control* subsection states the limitation explicitly: “Replacing each layer’s idiomatic path with identical SQL bounds how much of the spread is attributable to query strategy and formulation *jointly*; because the intervention changes several mechanisms together, it does not separate strategy from execution machinery.” Where I previously asserted the cause of Prisma’s slow raw path, the sentence is now labelled conjecture: “we conjecture this reflects dispatch and marshalling overhead in its raw API rather than SQL execution, a hypothesis the composite contrast cannot confirm.”

The Discussion paragraph *The same-SQL control* closes the loop: the contrast “bounds the combined contribution of query strategy and formulation without isolating a single factor such as eager-loading strategy,” and the Conclusion now reads that “running every layer on identical SQL narrows the spread substantially, but because that contrast changes several mechanisms together it does not attribute the difference to any single one.” The phrase “strategy gap” is gone; the residual is now the “idiomatic-to-same-SQL gap” throughout.

Prioritized checklist (P1–P10)

P1 — Same-SQL causal claim retracted (governing change)

Addressed in full above. The causal “strategy, not machinery” reading is withdrawn from both abstracts, the Results same-SQL subsection, the Discussion, and the Conclusion, and every residual is relabelled a bounded compound contrast. As one visible token of the rename, the Results now report that “MikroORM shows the largest idiomatic-to-same-SQL gap,” not a strategy gap.

P2 — "idiomatic" renamed to a documented, reproducible estimand

I no longer treat documentation order as evidence of production-representative use. The Study Design now states: “Documentation order is a reproducible selection heuristic, not evidence that the chosen API is performance-optimal or the most common production choice, so we report a documented default-configuration estimand rather than ‘the performance of each library used idiomatically’.” The no-maintainer-review disclosure is retained: “no evaluated library’s maintainers were involved or invited to inspect the adapters.”

P3 — Category-level recommendations scoped to the tested implementations

The practical guidance no longer generalizes to access-layer *categories*. It now reads: “the tested thin paths (the native driver and Knex) were the safer default here *for throughput and response-time p99*; whether that advantage generalizes to other implementations of those tiers is a hypothesis for broader study, not a category law,” and the paragraph opens by scoping the synthesis to “the benchmarked implementations in this configuration, workload, and host rather than to access-layer categories in general.”

P4 — Capacity, overload latency, and matched-utilization latency kept distinct

RQ1 now names the three quantities a fixed operating point conflates: “We keep distinct three quantities that a fixed operating point conflates: *capacity* (saturating throughput), *overload latency* (the saturated p99 under equal external demand), and *latency at matched utilization* (each layer driven at equal fractions of its own capacity).” The Study Design maps each to its experiment — equal external demand, equal utilization, and an equal compute budget — and the abstract reports all three operating conditions rather than one privileged point.

P5 — p99 tail count validated by a longer-window re-measurement

Because the primary p99 rests on roughly 60 tail observations per run, I added a 60 s re-measurement of the deep fetch. The Measurement procedure now states: “A 60 s re-measurement of the deep fetch (five times the requests) preserves the p99 ranking exactly on both engines and barely moves each cell’s p99 (Supplement Table S23), so the 12 s window is adequate for the tail comparison.” The full result is the new Supplement section *Longer-window tail sensitivity*; I give the numbers under Q4 below.

P6 — Adjacent-rank separation reported descriptively, not as confirmation

The significance prose is demoted to descriptive pairwise separation with the post-selection stated: “Because these adjacent pairs were selected after observing the ranking, we read their separation descriptively rather than as independent confirmation of the ordering, and it covers the deep fetch only (Table 6), not the full cross-pattern ranking.” The Analysis subsection likewise labels them “post-hoc adjacent-rank comparisons.”

P7 — Interaction reported as a magnitude, not only a p-floor

I added a supplement interaction-magnitude table. The main text keeps the honest caveat that the blocked-permutation p reaches its floor and “conveys little about magnitude,” and the new Supplement section *Layer-by-engine interaction magnitude* reports the size directly: “Supplement Table S28 reports *how large* the interaction is, as the PostgreSQL÷MySQL throughput ratio per layer and pattern with a 95% paired-bootstrap interval ... the reads hold a narrow band across layers (≈ 1.0 – 1.6) while the insert scatters from 1.6 to 3.2.”

P8 — Perishability tempered from a law to a version-sensitivity claim

Across all six sites the claim is now “can be version-sensitive,” anchored to one example rather than generalized. The Discussion heading reads “Access-layer rankings can be version-sensitive; the instrument is the contribution,” and the abstract states “Access-layer rankings can be version-sensitive, as one retired five-core result shows.” The Prisma case study is condensed to a single paragraph.

P9 — Length and repetition reduced

The condensed Prisma narrative and the removal of repeated restatements offset the new disclosures, so the body moved only marginally (11,376 to 11,388 words) while the abstract was tightened from 313 to 298 words; the new floats (longer-window tail, interaction magnitude, insert dispersion) live in the supplement only. Total main-text count under the IST rule is 14,934 words.

P10 — IST compliance

The structured abstract is 298 words (under 300); the total is 14,934 words (under 15,000); a top-level REPRODUCE.md was added; the raw-data checksums were regenerated to match the archived set 1:1; and the clean-room reproduction claim is verified (see Q7 and Q8).

§7 minor points

- **Intrinsic latency removed.** The term is gone everywhere; the tail now reads as a capacity effect, e.g. the saturated ladder “is a capacity-and-queueing effect, not a difference in tail latency at comparable utilization,” and the closed-loop tail is “the tail a deployment sees at saturation, not the per-request latency at moderate load.”
- **CRUD-spanning removed.** The endpoints are described as “five representative access patterns” in both the Introduction and the Study Design.
- **Latest-stable qualifier.** The abstract now pins “the latest stable library versions current at the July 2026 freeze,” matching the per-layer freeze rule in the Study Design and Threats.
- **Native leads all ten.** The claim is now the exact, verifiable “the native driver leads all ten engine-pattern combinations,” replacing “essentially every pattern.”
- **Vendor-independent qualified.** At first prominent use the Introduction reads “a single, vendor-independent benchmark (authorship only; Section 3 defines the term),” and the Study Design states “Vendor independence here means authorship only.”

- **Significance-table caveat.** The table is scoped to the deep fetch, “not the full cross-pattern ranking” (quoted under P6).
 - **Kendall’s τ added.** The abstract now reports “the read ordering is similar across engines (Spearman $\rho \geq 0.86$, Kendall $\tau \geq 0.71$),” and the supplement carries τ alongside every reported ρ .
 - **Insert dispersion.** A new supplement figure plots the raw per-replicate MySQL insert throughput: “several cells are visibly bimodal or heavy-tailed within a cell, structure a single CV cannot convey, which is why the insert rank correlation and dispersion are reported conservatively.”
 - **TOST reframed.** The Analysis now leads with the engineering consequence — “The $\pm 5\%$ equivalence margin is an author-selected smallest effect of practical interest for *this* study, not a general engineering threshold” — and demotes the noise coincidence: the margin happens to sit within the host’s run-to-run variation, “but that coincidence is not the justification.”
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The eight questions (Q1–Q8)

Q1 — Is documentation-first the same as idiomatic? No, and I no longer imply it. The documentation-order rule is a *reproducible selection heuristic*, not a claim of production-representative use; I renamed the estimand accordingly (P2) and rest its validity on the byte-equivalence cross-check and the archived per-adapter protocol (Supplement Table 16 and `METHODOLOGY.md`), which record, per layer, the exact API, its documentation page and version, and the alternative not used.

Q2 — What evidence identifies eager-loading as the dominant factor? None — that claim is retracted (P1). The same-SQL contrast is a compound bound that isolates no single mechanism, and the manuscript now says so at every site.

Q3 — Were the libraries’ maintainers asked to review the adapters? No, and this is disclosed as a limitation. Independence and external correctness review are compatible, but the latter was out of scope this cycle; the guarantee I do provide is the automated byte-equivalence cross-check that “asserts full JSON byte equality against the native-driver baseline, on both engines ... before any timing is collected.”

Q4 — How sensitive is the p99 to the run length? This is now answered in full, not pending. The new Supplement section *Longer-window tail sensitivity* re-measures the deep fetch over a 60 s window (about five times the requests, roughly 300 tail observations per run), ten independent runs, on both engines: “The p99 ranking is preserved exactly (Spearman $\rho = 1.00$ per engine), each cell’s p99 barely moves from the 12 s value, the p97.5 ordering matches the p99 ordering ($\rho = 1.00$), and the run-to-run p99 coefficient of variation stays small (maximum 6.0% on PostgreSQL, 3.8% on MySQL).” The 12 s window is therefore adequate for the tail *ranking*, and the tail conclusions do not rest on the exact per-run tail count (Supplement Table S23).

Q5 — Why is the saturated p99 the primary tail rather than matched utilization? The saturated p99 is deliberately an *overload* measurement, and I no longer privilege it implicitly. Following P4, the Study Design names equal external demand, equal utilization, and an equal compute budget as three co-primary operating conditions that “answer different questions and need not agree,” and the matched-utilization result is reported in the main text, where “the large saturated gap ... therefore largely dissolves; it is a capacity-and-queueing effect.”

Q6 — Do the conclusions survive excluding MikroORM? Yes. The Supplement *Per-layer cross-engine ranks* section recomputes the agreement over the remaining six portable layers: “the read ordering stays similar (deep-fetch $\rho = 0.77$, $\tau = 0.60$; range $\rho = 0.83$, $\tau = 0.73$) and the deep-fetch native-relative spread stays substantial at $3.7\times$ on both engines, so the RQ2 and RQ3 findings do not depend on that one layer.”

Q7 — Does the Zenodo archive regenerate every table clean-room? Yes. The new top-level `REPRODUCE.md`, section 4, gives the runnable clean-room steps: the immutable release is a `git` archive of the tagged commit, so it ships every tracked file including all 33 `results/*.json` raw-data files (force-added past `.gitignore`) and the committed generators, and the estimators are seeded so bootstrap intervals and permutation p-values are bit-reproducible. There is one disclosed caveat: “the round-trip-count table (Supplement S2) is derived from transient server statement logs that are not archived, so its committed `.tex` ... ships pre-generated rather than regenerable from the tarball; every other table regenerates from archived data.”

Q8 — Are all files in the immutable release? Yes. The archive tracks 33 `results/*.json` files (including `raw.json`), and `results/checksums.sha256` was regenerated to match that tracked set one-to-one — `sha256sum -c` over the tarball verifies every archived raw-data file, and the round-trip-count input is the single disclosed exception noted in Q7.

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Summary of changes

For the editor’s convenience: I retracted the same-SQL causal attribution and reworded it as a bounding contrast in both abstracts, Results, Discussion, and Conclusion (P1); renamed the treatment a documented default-configuration estimand and kept the no-maintainer disclosure (P2); scoped the practical guidance to the tested implementations (P3); separated capacity, overload latency, and matched-utilization latency in RQ1 and the Study Design (P4); validated the p99 tail count with a 60 s longer-window re-measurement, new Supplement Table S23 (P5, Q4); demoted the adjacent-rank significance to descriptive, post-selection, deep-fetch-only (P6); added a layer-by-engine interaction-magnitude table (P7); tempered perishability to a version-sensitivity claim with one condensed example (P8); trimmed length to 14,934 words with a 298-word abstract (P9, P10); added `REPRODUCE.md` and regenerated the raw-data checksums (P10, Q7, Q8); and completed the §7 minors, including the intrinsic-latency removal, the latest-stable and vendor-independent qualifiers, Kendall’s τ , the insert-dispersion figure, and the TOST reframe. I believe these revisions leave the manuscript claiming exactly what a single-host, current-version, default-configuration benchmark can support, and I am grateful that your interpretation concerns sharpened it. I look forward to your assessment.

Sincerely,

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